

No Pump-Free Corner: The Residual Doors of Covariant Modified Inertia, Computed — and a Pre-Registered Sign-Flip Signature for Dynamically Unsettled Galaxies

Carl P. Zimmerman (Briar Creek Tech) 2026-07-03 (v2 adds §6, the all-orders extension). Companion computations: 26 scripts, all exit 0, each lane independently and adversarially verified, at real_research/reviews/residual_doors_2026_07/ and real_research/reviews/fourth_order_2026_07/ in the supplementary repository.

Abstract

The six-theorem chain on covariant modified inertia (MI) — Ostrogradsky, Cassini/slip, the sign/state clause, the frequency law, the gas clamp, and the finite-time/non-stationary closure (Zenodo 10.5281/zenodo.21016309, 21139029, 21148494, 21152331, 21175723) — named its own residual doors rather than declaring the territory sealed. This note computes all four. **(D1) The interacting-field window** (the named mind-changer of Theorem VI): $F(\omega) \geq 0$ is a theory-independent Gram fact; a new monotone-window lemma extends Bisognano–Wichmann detailed balance to arbitrary window lengths on uniform trajectories for any interacting Wightman QFT; and a Källén–Lehmann reduction — independently re-derived on a genuine two-particle composite with zero fitted constants (agreement $1e-4$) — expresses the interacting vacuum’s windowed response on *any* trajectory as a positive mixture of free massive-slice responses, none of which invert (the sharp-kick margin asymptotes to zero from below; equality, never inversion). Driven interacting benchmarks do invert ($11.4\times$) — and the free control inverts $407\times$ under the same drive: agent-pumped Raman gain, the already-priced pump channel. Closed at 2nd order in detector coupling, the order at which the theorem’s object is defined; the honest residuals are 4-point effects and dS Bros–Moschella positivity. **(D2) The passive-anharmonic corner** — the last pump-free route: passivity *alone* (KMS not needed) forces $\delta m \geq 0$ in linear response at any anharmonicity, non-perturbatively in coupling for adiabatic worldlines (an exact term-by-term pair sum, re-derived independently to 0.58% with gap-sweep exponent -3.000000), with the degeneracy edge sealed. The adversarial verifier’s open-direction attack *partially succeeded* — instructively: at strong drive near the two-photon resonance ($gA=5$, $2\Omega \approx \text{gap}$) the reactive dressing genuinely goes negative from a passive start; the autopsy shows the drive de-passivizes the bath (the drive *is* the pump; Pusz–Woronowicz never violated), the window is frequency-locked, and band-mapping requires bath gaps $\lesssim 10^{-31}$ eV — the gas clamp. Amended closure: the passive softening channels are the sub-drive pole and its multiphoton/Floquet generalization, both drive-pumped, both clamped. **No pump-free MOND-sign channel exists.** **(D4) The flat-curve exceptional-point closure** of Theorem IV, done in full with Coriolis: the PT-broken band on flat curves is exactly $\mu \in (0.7579886, 0.8947874)$ — correcting our own published estimate 0.854, which sits *inside* the band — with a re-entrant real window (0.5, 0.758) the scalar shortcut missed; an exact fold law $\mu_{\text{fold}} = 1 - 1/(2p)$ shows sub-quadratic PU kernels ($p=1/2$) evade the EP entirely (deep-MOND PU orbits exist and are linearly stable — a genuine soft spot in the published Fourth Horn’s blanket claim),

and those kernels die instead by profile inversion (required vs delivered ($1-\mu$) slopes are sign-opposite; $\times 49-80$ on realistic disks). The horn stays dead; one of its bricks is relabeled in print. **(D3) The one door that opens:** fusing the Theorem-VI transient with the framework's memory-kernel σ -spread yields a pre-registered, dated prediction table for dynamically unsettled systems — Crater II +13.6–26.5%, Antlia II +3.4–6.7%, settled controls $<2\%$, and the clean MI-unique signature: **first-infall systems run 11–21% cold, flipping to hot after pericentre** — a relational sign flip that modified gravity (exactly zero by theorem) and tidal heating (no coherent deficit-then-excess) cannot mimic. Decisive at $3\sigma \approx 2029-2031$ (Gaia DR4 + WEAVE/4MOST dwarfs; CHANCES cluster infall populations). All lanes carry both coefficient footings ($a_0 = 9.36 \times 10^{-11}$ canonical vs 1.13×10^{-10}); no verdict flips. Verifier corrections are printed beside the results they amend, including two corrections to our own prior publications. **This version (v2) adds §6:** the two remainders D1 left open — 4-point/higher-order effects and the strict de Sitter light-field corner — are themselves computed, extending the sign wall from second order to *all orders* (Bisognano–Wichmann as an all-n-point algebra statement; exact mean-force Gibbs populations that never invert to $g=3$; a ground-state energy-conservation ledger showing every non-uniform-trajectory inversion is agent-work-funded, never free from the vacuum; and Euclidean-periodicity + stochastic-gap + band-separation for the de Sitter horizon), with named physics inputs rather than a single closed-form continuum proof, and with one self-caught overclaim corrected on the record.

1. Four named doors, four computations

A no-go program earns trust by naming its own edges and then walking to them. Theorem VI left four: one theoretical mind-changer (interacting fields), one pump-free corner (passive anharmonic baths), one bookkeeping debt (the flat-curve EP), and one phenomenological opening (unsettled populations). Method as throughout this program: independent compute lanes, each attacked by an adversarial verifier instructed to refute in *both* directions; every load-bearing number backed by a committed script (numpy/sympy, no network, exit 0); the recurring footing fork ($a_0 = cH_\Lambda/Z = 9.36 \times 10^{-11} \text{ m/s}^2$, ρ_{DE} , vs 1.13×10^{-10} , ρ_{total}/cH_0) carried everywhere.

2. D1 — the interacting-field window closes at second order

Theorem VI's positivity locks ($F(\omega) \geq 0$; $F(-\omega) - F(\omega) = (\omega/2\pi)\int \chi^2$) were proven for free fields pulled back to worldlines. Could an *interacting* vacuum, probed through a window on a non-uniform trajectory, invert?

Three closures and one honest benchmark:

1. **Positivity is universal.** $F(\omega)$ is the squared norm of a smeared field acting on the state — a two-line Gram argument valid for any state of any QFT. Confirmed to machine precision in a nonintegrable transverse-field Ising chain (static and moving probes; vacuum, excited, random, thermal states).
2. **Uniform trajectories, any interacting Wightman QFT.** Bisognano–Wichmann gives wedge-KMS regardless of interaction. A **monotone-window lemma** (new, proved, verified) then shows the detailed-balance inequality $F(\omega) \leq F(-\omega)$ survives *any* window length for

Gaussian-class windows: max deviation $\sim 2.7 \times 10^{-2}$ over the KMS-spectrum $\times \beta \times$ window-length scan, with the β -exponential recovered to five digits in the long-window limit. The lone counterfeit — boxcar windows can fake ratio ~ 23 by parking a sinc zero on a narrow line — vanishes for Gaussian windows ($\sim 10^{-118}$) and for all continua: a windowing artifact, not physics.

3. **Non-uniform trajectories.** Pusz–Woronowicz honestly does *not* close these (the external agent moving the detector supplies work — the Unruh effect itself excites detectors). So: compute. The load-bearing step is a **Källén–Lehmann reduction**: at 2nd order in detector coupling — the order at which Theorem VI’s F is defined — the interacting vacuum’s windowed response on *any* trajectory is a positive mixture of free massive-slice responses. The verifier re-derived this from scratch on a genuinely composite probe ($:\phi^2$; a two-particle continuum, KL density $\rho = 1/(\pi\mu^2\beta)$ obtained via the center-of-mass Jacobian) and confirmed the mixture against the direct position-space response on a non-uniform burst to $1-4 \times 10^{-4}$ **with zero fitted constants**. The free massive slices were then scanned over 28 burst/oscillation/kick configurations, and adversarially over 7 more beyond the grid (light masses, sharp fast bursts, chirps, off-center windows, resonance-tuned oscillations): **no inversion anywhere**. The sharpest kicks drive the no-inversion margin monotonically toward zero *from below* ($-4.6 \times 10^{-2} \rightarrow -1.9 \times 10^{-2}$): the sudden limit approaches equality and never crosses — in 3+1D the $k^2 dk$ phase space never blocks the emission resonances that a discrete lattice can.
4. **The driven benchmark keeps the bookkeeping honest.** A nonperturbatively driven interacting lattice (Kim–Huse class, $L=9$) *does* invert — $F(\omega)/F(-\omega) = 11.4$ at the drive-Stokes sideband $\omega^* = 2\Omega_{\text{drive}} - \Delta E$. But the free-fermion control under the identical drive inverts **407×**. Inversion is not interacting-specific; it is agent-pumped Raman gain — precisely the pump channel the chain already prices at $\times 2.9 \times 10^{10}$ shortfall.

Verdict: CLOSED at 2nd order. The two remainders this left — 4-point/4th-order effects, where the KL reduction is silent, and Bros–Moschella dS positivity for light interacting fields — are themselves computed in §6 (the all-orders extension added in this version).

3. D2 — passivity alone forces the anti-MOND sign: the pump requirement is fully general

The last pump-free hope: Theorem III’s $\delta m \geq 0$ was proven for harmonic/two-level passive baths; anharmonic baths escape *state-blindness* — could a passive-but-anharmonic medium carry $\delta m < 0$ with no power source?

No — and now at theorem grade:

1. **Linear response, any anharmonicity, any stationary passive state.** In the joint ρ/H eigenbasis every Bohr-line weight is $(p_n - p_m)|B_{nm}|^2 \geq 0$ under the passivity ordering, so $\omega \cdot \text{Im}\chi(\omega) \geq 0$, and the worldline dressing $\delta m = (2/\pi)g^2 \int \text{Im}\chi/\omega^3 \geq 0$. **KMS is not needed** — passive-but-not-KMS states are covered directly. (Minimum Bohr-line weight over 2000 random anharmonic passive ensembles: $+1.2 \times 10^{-8}$; over the passive polytope’s extreme points: 0 exactly, term-by-term.)
2. **Non-perturbative in coupling** (adiabatic worldlines): the $O(v^2)$ mass coefficient is the exact pair sum $2g^2 \sum (p_n - p_m) |\langle m | \partial H / \partial x | n \rangle|^2 / \omega^3$ over the *full displaced* spectrum — term-by-term

≥ 0 for states passive with respect to the instantaneous Hamiltonian, a premise the verifier tested directly (exact slow sweep: zero ordering violations along the trajectory; finite-speed reordering is Landau-Zener work, i.e. drive-paid). Exact-diagonalized quartic scans across coupling, displacement, and passive-state families: global minimum $+3.2 \times 10^{-4}$, while the most adversarial *non*-passive state reaches -0.0038 — passivity, not model structure, is what binds. The lane caught and killed its own fake counterexample (a Fock-truncation artifact printing $\delta m = -213$; converged truncation $\rightarrow$ all-passive positive). The verifier re-derived the pair-sum law on a different anharmonic model, truncation, and integrator: direct dynamical mass measurement agrees to 0.58%, gap-sweep log-log slope -3.000000 , harmonic anchor to 6×10^{-17} .

3. **The degeneracy edge** — passivity permits arbitrary coherences inside degenerate subspaces, the sharpest crack candidate: such coherences commute with the block Hamiltonian and contribute constant geometric forces with *zero* inertia (force-variation 9×10^{-13}); 300 coherent passive states all harden (min $+0.52$); near-degenerate passive pairs harden as $+1/\omega_0^3$.

The verifier’s catch — an open-direction attack that partially succeeded, and what it teaches. Beyond the lane’s scan range (drive amplitude $gA=5$ at the two-photon resonance $2\Omega \approx \text{gap}$), the reactive dressing **genuinely goes negative from a passive start** ($\delta m = -0.0169$; -0.007 from thermal $T=2$). The lane’s “never crosses zero” was a scan-range artifact — corrected in print here. The autopsy is the point: (i) the effect is frequency-locked to a narrow window ($\Omega/\text{gap} \in [0.40, 0.56]$; positive again by 0.60) — a *frequency* law, never $\mu(a/a_0)$; (ii) the bath *absorbs* energy throughout ($+5.07$ per protocol; Pusz–Woronowicz never violated to $gA=5$); (iii) the drive de-passivizes the bath before softening appears (6 population-ordering violations vs 0 initially) — **the drive is the pump**; (iv) mapped to the galactic band it requires bath gaps $\lesssim 10^{-31}$ eV, where the gas clamp (Theorem V) applies exactly as it does to the sub-drive pole.

Amended closure (the citable sentence): passivity alone forces $\delta m \geq 0$; the only passive-start softening channels are the sub-drive pole *and its multiphoton/Floquet-dressed generalization* — both frequency-locked, both drive-pumped, both clamped by Theorems IV/V. **There is no pump-free MOND-sign channel.** Footing-independent (spreads printed; 20.8% on a_0 , no flips).

4. D4 — the flat-curve EP, exactly: a published number corrected, a soft spot found, the horn dead by a different brick

Theorem IV (Zenodo 10.5281/zenodo.21148494) proved its exceptional-point cap $\mu \geq 1/2$ harmonic-exactly, and carried a crude flat-curve estimate ~ 0.854 “pending the Coriolis linearization.” Done in full — the radial–azimuthal PU linearization around a circular orbit in the self-consistent flat-curve potential ($\kappa = \sqrt{2} \cdot \Omega$), degree-8 characteristic polynomial, cross-checked by companion-matrix eigenvalues *and* nonlinear orbit integration (measured runaway $0.296 \cdot \Omega$ vs 0.300 predicted, 1.5%):

- **Exact flat-curve band:** PT breaks for $\mu \in (0.7579886, 0.8947874)$ — the roots of $664\mu^4 - 3036\mu^3 + 5086\mu^2 - 3681\mu + 971$. The published 0.854 sits *inside* the band: wrong in both directions. There is a re-entrant PT-real window (0.5, 0.758) the scalar treatment missed; below $\mu = 1/2$ the spectrum is doubly unstable (real saddle + complex quartet), so deep-MOND orbit non-existence is *confirmed for the quadratic proxy*. Family map across rotation-curve shapes $\kappa/$

$\Omega \in [1, 2]$: the upper band edge is nearly universal (0.892–0.9); the harmonic case is the unique never-breaking endpoint — explaining retroactively why the harmonic-exact cap missed the band entirely.

- **The fold law and the genuine soft spot.** For general PU kernels $k(y)$, the fold obeys the exact factorization $\mu_{\text{fold}} = 1 - 1/(2p)$. Sub-quadratic kernels — including the framework-tail case $p = 1/2$ — are **PT-real at all $\mu < 0.5486$** : deep-MOND PU orbits *exist and are linearly stable* there. The Fourth Horn’s blanket “deep MOND fails by orbit non-existence” was quadratic-proxy-specific — a correction to our own published theorem’s *mechanism*, found by this program’s both-ways rule and printed here. (The verifier independently confirmed the fold law nonlinearly at $p = 3/4$ — fold exactly $\mu = 1/3$ — and confirmed the soft spot is not a degeneracy artifact.)
- **Why the horn stays dead:** those evading kernels force $d \ln(1-\mu)/d \ln R = -2p$ along a flat curve, while the framework’s own v requires $\approx +1$ — sign-opposite slopes, a mismatch of **$\times 49$ – 80 on realistic 2–20 kpc disks** ($p=1/2$; $\times 493$ – 799 for $p=1$; verifier-corrected from the $\times 100$ asymptotic bound; $\sim 10\%$ footing spread, no flips). Profile inversion does the work the EP was credited with.
- **Scope:** the PU horn only. The framework’s own nonlocal-kernel MI and its phenomenology are untouched by this lane.

5. D3 — the door that opens: a pre-registered sign-flip signature for unsettled galaxies

Theorem VI quantified a residual: one-shot transients contribute at the percent level per decade of orbits — invisible in settled disks (the RAR), potentially distinctive where relaxation is recent. Fused here with the framework’s banked memory-kernel physics (Lorentzian kernel, $\tau = 0.45$ Gyr, $\theta_0 \in [\sqrt{2}, 2]$) into one amplitude-vs-settledness observable, computed on the framework’s own objects, with the corpus’s own orbit machinery, and corrected by adversarial verification (four corrections folded; the two-channel double-count and a factor-2 print bug removed — the table below is the defensible, memory-channel-only version).

The pre-registered table (relational σ signal vs settled expectation; both footings, no flips):

System / class	Predicted signal	Phase context
Crater II	+13.6 to +26.5%	post-pericentre ($t_{\text{peri}} = 0.77$ Gyr, phase 0.27)
Antlia II	+3.4 to +6.7%	near apocentre (phase 0.96)
Sagittarius	+23 to +27%	tidally dirty — excluded as a clean carrier
First-infall dwarfs / cluster members	-11 to -21% (deficit)	before first pericentre — the sign flip
Leo I	≈ 0	density-gate null control
Settled controls (Fornax/ Sculptor class)	< 2%	
Cluster relational spread (recent-infall vs virialized)	+6 to +13%	recovers the banked MI band

The clean MI-unique signature is the sign flip, not the decay clock: the verifier showed the $\tau \approx 0.45$ Gyr exponential decay is degenerate with post-shock tidal revirialization on diffuse carriers ($t_{\text{cross}} = 0.86 \cdot \tau_{\text{mem}}$ for Crater II — structurally forced for $y \sim 1$ carriers). What survives is stronger: **a coherent σ deficit on first infall that flips to an excess after pericentre**. Tidal heating cannot produce a coherent pre-pericentre deficit; modified gravity and instantaneous- μ MI predict exactly zero relational effect (Milgrom’s MG-virial universality); Λ CDM+tides produce excess-only with hysteresis. The Luo window tail itself is restated honestly as a one-sided $\leq$ envelope (0.4–4.0% for $n_{\text{orb}} = 10 \rightarrow 1$) — no standalone test, since a null cannot falsify an envelope.

Decisiveness: matched-pericentre, matched-density phase comparisons. MW-dwarf arm: ~ 9 carriers/arm at 10% signal and 7% σ -error $\rightarrow 3\sigma \approx$ **2029–2030** (Gaia DR4 orbital phases, Dec 2026, + WEAVE/4MOST velocity dispersions). Cluster arm: 6% signal vs 2% bin error $\rightarrow 3\sigma \approx$ **2028–2031** (CHANCES recent-infall vs virialized populations). Kill conditions, stated now: DR4 revising Crater II/Antlia II orbital phases beyond ~ 2 Gyr post-pericentre kills the carrier liveness; a computed kernel shrinking the θ band below 5% kills the amplitude; tidal simulations producing a coherent *pre-pericentre deficit* with a pericentre flip would destroy the signature’s uniqueness.

6. The interacting window, extended: the sign wall is all-orders (*added in v2*)

D1 closed the interacting window *at 2nd order in the detector–field coupling g* — the order at which Theorem VI’s response function F is defined — and named the two remainders honestly: 4-point/higher-order effects (where the Källén–Lehmann reduction is silent) and the strict de Sitter complementary-series positivity question. Those are the *last computable* places the MOND sign could hide in the interacting vacuum with no pump. We computed them, on three fronts (all scripts exit 0, `real_research/reviews/fourth_order_2026_07/`; ledger `real_research/`

FOURTH_ORDER_SIGN_WALL_2026-07.md). The wall holds — effectively to all orders — with named physics inputs rather than a single closed-form continuum proof.

Uniform (KMS) trajectories — the lock is all-orders. Bisognano–Wichmann is an *algebra* statement: the Minkowski vacuum restricted to the Rindler wedge is KMS at T_U for the *entire* n -point hierarchy of any interacting Wightman QFT, not merely the 2-point function — so a detector coupled at any strength thermalizes toward T_U at every order. Two exact, coupling-nonperturbative checks confirm it: (i) the KMS detailed-balance relation $S(-\omega) = e^{-\beta\omega}S(\omega)$ holds to 6.6×10^{-13} for a genuinely quartic (interacting) bath, fixing the up/down rate ratio below unity at every order; (ii) the exact **mean-force Gibbs state** of a qubit strongly coupled to an anharmonic bath — the true all-orders stationary state, no Born, no RWA — never inverts: the bare-basis gap $p_g - p_e$ stays positive across $g = 0 \rightarrow 3$, minimum **+0.09 at $g=3$** , where mean-force has already displaced populations by 0.41 from bare Gibbs. The one candidate crack (mean-force shifting bare-basis populations) shifts the gap but never flips its sign.

Non-uniform trajectories — exact, all orders automatically. Off uniform motion there is no wedge and the external agent does work; the 2nd-order KL lock is silent. We evolved a detector dragged on non-uniform protocols through the *exact ground state* of a nonintegrable mixed-field Ising chain (genuine connected 4-point functions), at strong g (exact unitary evolution contains every order). The decisive invariant is exact and needs no adiabatic assumption: **the chain starts in its ground state — the global energy minimum — so it can only absorb energy, never donate it.** Hence $E_{\text{chain_gain}} \geq 0$, and by energy conservation $E_{\text{det}} = W_{\text{agent}} - E_{\text{chain_gain}} \leq W_{\text{agent}}$: every detector excitation or inversion is paid for by the moving agent’s work, never by the vacuum (verified per run; $E_{\text{chain_gain}} = 0.24, 0.72, 3.85$ across the three protocols). Two corollaries: the interacting chain absorbs the agent’s work *more* effectively than the free chain (4-point structure aids the **pump**, never the vacuum — the opposite of a free sign), and the work-free (adiabatic) limit leaves the detector un-inverted (the exact dressed ground state has excited weight $0.33 < \frac{1}{2}$, i.e. $\delta m \geq 0$).

An overclaim caught and corrected, on the record. A first pass labeled a single monotone kick a “vacuum probe” and reported it did not invert. Its own robustness sweep (54 configurations) refuted this — sudden single kicks *do* invert (excited weight up to 0.83) because a sudden switch-on is itself a quench that does work. The probe/drive distinction was physically empty; it was replaced by the exact energy-conservation ledger above, which is *stronger* (no adiabatic assumption). The correction is retained here per this program’s practice of recording kills and corrections with the same weight as confirmations.

The de Sitter corner. For the framework’s actual (de Sitter) bath, the open edge was interacting *light* fields (complementary series), where perturbative IR behaviour and dS Källén–Lehmann positivity are unresolved. Two computable bounds close it for the framework’s purposes: (i) an interacting theory periodic in Euclidean time $\beta_{\text{dS}} = 2\pi/H$ inherits KMS *independent of coupling* (verified to 1.9×10^{-12} in the thermal-circle proxy across $\lambda = 0 \rightarrow 4$) — the Bros–Epstein–Moschella Euclidean construction, whose only obstruction is IR-existence for light fields; (ii) the Starobinsky–Yokoyama stochastic mass $m_{\text{eff}} \approx 0.36 \lambda^{1/4} H$ renders interacting light fields IR-finite, and in any case the framework’s inertia response is **in-band at $\omega/H \in [15, 10800]$ (canonical) /**

[15, 8900] (**alternate**) — every relevant mode has $\omega \gg H$, 1–4 orders above the $\omega \lesssim H$ region where any complementary-series subtlety could live. The unresolved sliver cannot source in-band inertia.

What §6 establishes. The sign wall now runs to all orders on every route we can compute: free fields (state-blind, all times), stationary passive baths (any anharmonicity, §3), interacting vacua at 2nd order (§2) *and beyond* (uniform: BW all-orders + mean-force; non-uniform: ground-state-passivity ledger), and the interacting de Sitter horizon. Interactions make pumping *easier*, never free — they do not help the framework. The residual, named: a fully rigorous continuum 4-point theorem (we have BW modular theory, exact lattice/ledger arguments, and named physics inputs, not one closed-form all-orders continuum proof), and the strict complementary-series dS positivity question in the mathematical literature (bypassed here by band-separation and the stochastic gap, not resolved in general).

7. Standing

- **The pump requirement is now fully general** (D2): no passive medium of any structure, probed any way — stationary, transient, windowed, driven-from-passive — carries the MOND sign without a pump, and every named pump is priced and clamped. Combined with D1, the sign wall now runs: free fields (state-blind, all times) $\rightarrow$ structured passive baths (theorem, all anharmonicities) $\rightarrow$ interacting vacua (positive-mixture reduction, 2nd order) $\rightarrow$ pumped channels (priced $\times 2.9 \times 10^{10}$, gas-clamped).
- **Theorem IV’s flat-curve bookkeeping is exact** (D4), its published estimate corrected, its conclusion re-derived through profile inversion — the kill survives its own mechanism’s correction.
- **The framework gains one new dated, falsifiable, MI-unique observable** (D3): the first-infall sign flip, decisive ~2029–2031 — joining $\bar{s}^{\wedge}TX$ (10.5281/zenodo.21137568), Gaia DR4 wide binaries, and DESI DR3 on the live ledger.
- **Doors that remain open, named:** a fully rigorous *continuum* 4-point theorem (§6 gives Bisognano–Wichmann + exact lattice/ledger arguments + named physics inputs, not one closed-form all-orders proof); the strict complementary-series dS positivity question (bypassed in §6 by band-separation and the stochastic gap, not resolved in general); the D3 table awaiting data; the empirical fronts. Per this program’s standing rule, there is no “theory closed” — there is a map with computed edges.

8. Verification and scope

Every lane was adversarially verified by an independent agent instructed to attack in both directions; two verifier runs died on infrastructure and were re-run (their targets’ scripts all execute in <10 s, exit 0). Corrections are printed beside the results they amend: D1’s number-provenance slip and grid-limitation sharpened (margin $\rightarrow 0^-$, never crossing); D2’s “never crosses 0” corrected by a partially-successful attack whose autopsy *strengthened* the closure (the Floquet channel named and clamped); D3’s factor-2 bug, envelope double-count, and decay-clock degeneracy removed (the printed table is the corrected one); D4’s asymptotic $\times 100$ margin brought down to the realistic $\times 49$ –80. The §6 all-orders extension carries its own self-caught correction (the “vacuum probe” mislabel, refuted by its own robustness sweep and replaced by the exact ground-state ledger).

Three of these amend our *own* prior work (the Fourth Horn’s flat-curve estimate and its orbit-non-existence mechanism; and §6’s first-pass framing) — reported here with the same weight as the wins.

Scope: this note computes the named residual doors of the six-theorem chain. It does not claim “modified inertia is impossible” beyond the mapped territory; it does not derive a_0 ’s value or the MOND sign (the sign requires a pump; no candidate exists); and it does not touch the $a_0 = c^2\sqrt{(\Lambda/32\pi)}$ reframing or its live empirical tests. The author retracted all earlier theory-of-everything claims (2026-06-23); this note claims computed edges and one new dated prediction, not a completion.

Script index

Door	Scripts (real_research/reviews/ residual_doors_2026_07/, all exit 0)
§6 all-orders (.../fourth_order_2026_07/)	L1_meanforce_allorders.py, L2_nonuniform_interacting_vacuum.py, L2b_adiabatic_dressing.py, L3_dS_interacting_corner.py, verify_L2_vacuum_probe_robustness.py
D1	D1_1_general_positivity.py, D1_2_kms_window_pw_theorems.py, D1_3_kl_reduction_burst_scan.py, D1_4_interacting_lattice_benchmark.py, verify_D1_interacting_window_rederive.p
D2	D2_1_theorem_passivity_dispersion.py, D2_2_quartic_nonperturbative.py, D2_3_degeneracy_edge.py, verify_D2_passive_anharmonic_rederive.py, verify_D2_rederive_final.py
D3	D3_amplitude_vs_settledness.py, D3_named_systems_forecast.py, D3_discrimination_detectability.py, verify_D3_unsettled_observable_rederive.p
D4	D4_flat_ep_cap.py, D4_curve_family.py, D4_tail_kernel_kill_margin.py, verify_D4_flatcurve_EP_rederive.py

Ledger: real_research/RESIDUAL_DOORS_VERDICT_2026-07.md.

References (chain)

- Zimmerman, C.P., *Scale Without Law*, 10.5281/zenodo.21016309
- Zimmerman, C.P., *The Sign Premise Is a State Clause*, 10.5281/zenodo.21139029 (Theorem III)
- Zimmerman, C.P., *The Fourth Horn*, 10.5281/zenodo.21148494 (Theorem IV — flat-curve estimate corrected herein)
- Zimmerman, C.P., *The Kernel That Builds Its Own Laser*, 10.5281/zenodo.21152331 (Theorem V)

- Zimmerman, C.P., *Scale Yes, Shape Yes, Sign No*, 10.5281/zenodo.21175723 (Theorem VI — residual doors named)
- Zimmerman, C.P., *Orbital History Writes the Cosmic Clock on Dwarf Galaxies* (v3), 10.5281/zenodo.20969585 (the memory kernel)
- Luo, M.J., arXiv:2602.14515 (2026)
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